



FINAL e-Learning System Flow (Corrected & Complete)

1. Admin Flow (Chee Luo)

A. Admin Account Management

- Admin logs in
- Admin dashboard overview
- Manage teacher accounts (create/suspend/reset password)
- Manage student accounts (students self-register, admin supervises)

B. Category & Subject Management

- Create/edit/delete categories
- Assign teachers to categories
- Approve or reject teacher course creation requests

C. Payment & Enrollment Management

- View enrollments
- View transactions
- Manage promo codes / discounts

D. Course Governance

- View all courses
- Publish / unpublish courses
- View course statistics
- Monitor teacher assessments (optional)

E. System Settings

- Certificate template management
 - Content storage settings
 - SMTP / notification email settings
 - Platform branding (logo, color theme)
 - Role & permission settings
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2. Teacher Flow (Mocha)

A. Teacher Dashboard

- My Courses (ViewCourse)
- Add New Course (CreateCourse)
- Assessments (ViewAssessment , CreateAssessment)
- Students Enrolled  (Studen/CoursetManagement)
- Course Analytics  (Studen/CoursetManagement)

B. Course Creation (CreateCourse)

- Create course (title, category, description, thumbnail, difficulty)
- Upload materials: PDFs, videos, text lessons
- Add coding demos
- Save course as draft
- Submit course for admin approval

C. Assessment Creation (CreateAssessment)

- Select course
- Add MCQs
- Set correct answer
- Set passing mark
- Preview & publish assessment

D. Student / Course Management

- View enrolled students
 - Track student progress
 - View assessment results
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3. Student Flow (Ivya)

A. Browsing & Purchasing

- Student self-registration & login
- Browse course catalog
- Filter/search courses
- View course preview, syllabus, teacher info, price
- Checkout → Payment
- Enrolled course appears in My Courses

B. Learning

- Access course content
- Watch videos
- Read notes
- Download materials
- Track progress

C. Assessment

- Attempt assessment
 - Auto-grading
 - View score
 - Get certificate if passed
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4. Auto-Grading Flow (Yee Xian)

- Student submits assessment
 - System fetches correct answers
 - Compare answers
 - Calculate score
 - Pass/Fail decision
 - Store score in database
 - Update course progress
 - If PASS → trigger certificate generation
 - Record attempt history
 - Notify student (email or dashboard notification)
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5. Certificate Flow (Jia)

- System creates unique certificate ID
- Generate PDF certificate
- Include QR code with verification link
- Save certificate path to database
- Show in "My Certificates" page
- Certificate verification page for QR code scanning

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FINAL DELIVERABLE — EXACT LIST OF WHAT EACH PERSON MUST BUILD

Below is the exact breakdown you asked for:

- ✓ Front-end pages
 - ✓ Back-end APIs
 - ✓ Database tables
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1. CHEE LUO — Admin Module Lead

Build These Pages

- Admin Login Page
- Admin Dashboard
- Manage Teachers Page
- Manage Students Page
- Categories Management Page
- Course Approval Page
- Transactions Page
- Promo Code Page
- System Settings Page

Build These APIs

- `/api/admin/create-teacher`
- `/api/admin/manage-student`
- `/api/admin/categories`
- `/api/admin/approve-course`
- `/api/admin/transactions`
- `/api/admin/settings`

Build/Use These Database Tables

- Users (admin, teachers, students)
- Categories
- Courses
- Transactions
- PromoCodes
- Settings



2. MOCHA — Teacher Module Lead

Build These Pages

- Teacher Login
- Teacher Dashboard
- Create Course
- Upload Materials
- Create Assessment
- View Enrolled Students
- Course Analytics

Build These APIs

- `/api/teacher/create-course`
- `/api/teacher/upload-material`
- `/api/teacher/create-assessment`
- `/api/teacher/my-courses`
- `/api/teacher/student-progress`

Build/Use These Database Tables

- Courses
 - CourseMaterials
 - Assessments
 - Questions
 - TeacherProfiles
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3. IVYA — Student Module Lead

Build These Pages

- Student Registration
- Student Login
- Course Catalog
- Course Details
- Payment Page
- My Courses
- Lesson Viewer
- Assessment Attempt Page

- My Certificates

Build These APIs

- `/api/student/register`
- `/api/student/enroll`
- `/api/student/progress`
- `/api/student/attempt-assessment`
- `/api/student/certificates`

Build/Use These Database Tables

- Users
 - Enrollment
 - Transactions
 - StudentProgress
 - StudentWishlist
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4. YEE XIAN — Auto-Grading Lead

Build These Services/APIs

- `/api/grading/submit` (core grading logic)
- `/api/grading/result`
- `/api/grading/attempt-history`
- Auto-trigger certificate generation

Build These Backend Components

- MCQ comparison logic
- Calculate score
- Pass/Fail logic
- Timer logic
- Shuffling algorithm

Build/Use These Database Tables

- StudentAnswers
 - Scores
 - AssessmentAttempts
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5. JIA — Certificate System Lead

Build These Pages

- Certificate Viewer
- QR Verification Page

Build These APIs

- `/api/certificate/generate`
- `/api/certificate/verify`
- `/api/certificate/history`

Build These Backend Components

- PDF generator
- QR code generator
- Certificate file storage

Build/Use These Database Tables

- Certificates
- CertificateTemplates